

RAMYA AKULA, Ph.D.

San Jose, California, USA

Machine Learning Researcher | Trustworthy AI | Computational Biology | Natural Language Processing

[Google Scholar](#) (267 Citations | 9 h-index | 9 i10-index) | [LinkedIn](#) | ✉ ramya.akula01@gmail.com

RESEARCH IDENTITY

I develop trustworthy machine learning methods for understanding complex biological systems and human language, with the long-term goal of improving the reliability of AI in healthcare.

RESEARCH THEMES

Natural Language Processing	Computational Biology	Trustworthy AI
Large Language Models, Figurative Language, Text Summarization, Semantic Representation Learning	Single-cell Transcriptomics, Lung Aging, Cellular Senescence, Machine Learning for Biology	Evaluation methodologies, Interpretability, Trustworthy AI, Human-AI Collaboration

Research Recognition: DARPA Forward Riser (2022), U.S. Patent, NSF I-Corps Entrepreneurial Lead.

- **Scientific Contributions:** Developed an interpretable self-attention architecture for modeling semantic ambiguity in figurative language. Proposed SPEED Score, a semantic evaluation metric for abstractive and extractive summarization. Developed scalable computational pipelines for integrating multi-cohort single-cell transcriptomic datasets. Applied machine learning to characterize aging, senescence, and cellular heterogeneity in lung tissue.

SELECTED PUBLICATIONS

- **Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection in Social Media**, *Entropy* (2021)
- **SPEED Score: Sentence Pair Embeddings Based Evaluation Metric for Abstractive and Extractive Summarization**, *Language Resources and Evaluation Conference (LREC)* (2022)
- **Explainable Detection of Sarcasm in Social Media**, *ACL Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis* (2021)
- **Ethical AI for Social Good**, *International Conference on Human-Computer Interaction* (2021)

RESEARCH CONTRIBUTIONS & FUTURE DIRECTIONS

Developing reliable and interpretable machine learning methods for high-consequence applications spanning natural language processing, computational biology, and healthcare.

- **NLP:** Interpretable transformer models and evaluation methods for figurative language and summarization (e.g., Interpretable Multi-Head Attention, SPEED metric, explainable sarcasm detection). Outcomes: DARPA Forward Riser, U.S. patent application, publications (ACL, LREC, Entropy, HCI).
- **Computational Biology:** Reproducible pipelines for multi-cohort single-cell RNA-seq analysis (Harmony integration, cell-type annotation, differential expression) applied to lung aging and senescence. Peer-reviewed publication.

Future Work: Trustworthy clinical AI—evaluation of conversational systems, adversarial testing of LLMs, and uncertainty communication, integrating ML, NLP, and computational biology.

EDUCATION

- University of Central Florida
 - Ph.D., Computer Science (2018–2022)
 - Advisor: Dr.Ivan Garibay
 - Dissertation: *Toxic and Figurative Language Detection and Evaluation Metrics for Social Media Content*
- Technical University of Kaiserslautern, Germany
 - M.S., Computer Science (2013-2015)
 - Advisor: Dr. Hans Hagen
 - Dissertation: Hierarchical Visualization of Complex Social Networks.
- Jawaharlal Nehru Technological University, India
 - B.Tech., Computer Science (2006-2010)
 - Dissertation: Chat Client Application with Multi-thread Execution.

ACADEMIC APPOINTMENTS & EXPERIENCE

- **The Ohio State University**
 - **Postdoctoral Scholar** | Department of Internal Medicine (Pulmonary)
 - Advisor: Dr. Mauricio Rojas
 - **Research Focus:** Single-cell transcriptomics, Machine learning for computational biology, Lung aging and cellular senescence, Reproducible computational workflows
- **University of Central Florida**
 - **Lecturer** | Computer Science & FinTech Program
 - Designed and taught graduate and undergraduate courses in Artificial Intelligence, Natural Language Processing, Machine Learning, Data Science, Algorithms and Blockchain in Masters in FinTech program while continuing an active research program. Supervised graduate capstone projects and mentored undergraduates.
 - **Graduate Research Assistant** | Department of Computer Science
 - Advisor: Dr. Ivan Garibay
 - Research Focus: Natural language processing, Explainable AI, Deep learning, Trustworthy machine learning, Language evaluation metrics
- **Technical University of Kaiserslautern(RPTU)**
 - **Graduate Research Assistant** | Department of Computer Science
 - Advisor: Dr.Hans Hagen
 - Research Focus: Machine Learning, Visualization of complex social networks.
- **Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI) GmbH**
 - **Graduate Research Assistant** | Department of Computer Science
 - Advisor: Dr.Didier Stricker
 - Research Focus: Machine Learning, Intelligent Systems

Duration	Role	Organization	Department / Details	Location
2024 – 2025	Postdoctoral Scholar	The Ohio State University	Dept. of Internal Medicine, Pulmonary	USA
2023 – 2024	Lecturer	University of Central Florida	MS in Fin-Tech, Dept. of Computer Science	USA
2018 – 2022	Graduate Research Assistant	University of Central Florida	Dept. of Computer Science	USA
2015 – 2017	Project Engineer	Umlaut GmbH (formerly P3 Group)	Client: Airbus GmbH	Germany

2014 – 2015	Graduate Research Assistant	Technical University of Kaiserslautern-Landau (RPTU)	Dept. of Computer Science	Germany
2013 – 2014	Software Quality Assurance Analyst	Exco GmbH	Client: Roche GmbH	Germany
2013	Graduate Research Assistant	DFKI GmbH	Artificial Intelligence Research	Germany
2011 – 2013	Software Engineer	Osmania University	Urban & Environmental Studies	India
2010 – 2011	Software Developer	Inventurus Software Solutions	Client: United Health Care	India

SELECTED AWARDS & RECOGNITION

- **DARPA Forward Riser (2022)**
- Selected as a DARPA Forward Riser in recognition of innovative research in interpretable machine learning for figurative language understanding.
- **U.S. Patent (2025)**
 - **Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection.** Patent application arising from doctoral research on interpretable neural architectures. US Patent 12,210,833, 2025
- **NSF I-Corps Entrepreneurial Lead(2019)**
 - Selected to lead customer discovery and commercialization activities for an AI-assisted qualitative analysis platform funded through the National Science Foundation.
- **Startup Weekend Orlando – First Place(2021)**
 - Recognized for developing an AI-powered online youth safety platform.

SPONSORED RESEARCH

- **DARPA SocialSim:** Developed interpretable language understanding methods for analyzing complex social media interactions.
- **DARPA ASIST:** Applied machine learning techniques to improve AI-assisted decision support for collaborative environments.
- **NSF I-Corps:** As Entrepreneurial Lead, translated AI research into customer discovery, commercialization strategy, and technology validation.

TEACHING & MENTORSHIP

- **Graduate:** Advanced Natural Language Processing, Artificial Intelligence in FinTech, Machine Learning Applications, Algorithms, Blockchain and Smart Contracts, Data Science Programming
- **Undergraduate:** Introduction to C, Data Analytics, Programming Foundations
- Mentored 60 graduate and 20 undergraduate students in machine learning and data science projects. Guided student teams in applied AI research and software development.

TECHNICAL EXPERTISE

- **Machine Learning:** Deep Learning, Transformer Architectures, Attention Mechanisms, Transfer Learning, Representation Learning, Explainable AI

- **Natural Language Processing:** Figurative Language Understanding, Text Summarization, Semantic Similarity, Toxic Language Detection, Large Language Models
- **Computational Biology:** Single-cell RNA Sequencing, Seurat, Harmony, Differential Expression, Cell Annotation, Gene Set Enrichment, Transcriptomics
- **Statistical & Computational Methods:** Machine Learning Evaluation, Reproducible Computing, Statistical Inference, High-Dimensional Data Analysis, Python, R, Jupyter
- **Scientific Computing:** High-Performance Computing (HPC), Git, Linux, Docker

SCHOLARLY IMPACT & OUTREACH

Peer-Reviewed Journal Articles

- **Akula, R., et al. Transcriptomic shifts in human lungs donated after brain and circulatory death and severe Obliterative Bronchiolitis.** *The Journals of Gerontology*, 2026.
- Pinky, J. N., & Akula, R. (2025). **Enhancing cryptocurrency market forecasting: Advanced machine learning techniques and industrial engineering contributions.** *Advances in Artificial Intelligence Applications in Industrial and Systems Engineering*, 197-262.
- **Akula, R., Garibay, I. Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection in Social Media.** *Entropy*, 2021.
- **Akula, R., Garibay, I. Viztract: Visualization of complex social networks for easy user perception.** *Big Data and Cognitive Computing*, 2018.

Refereed Conference Proceedings

- **Akula, R., Garibay, I. SPEED Score: Sentence Pair Embeddings Based Evaluation Metric for Abstractive and Extractive Summarization.** *LREC*, 2022.
- **Akula, R., Garibay, I. Explainable Detection of Sarcasm in Social Media.** *ACL Workshop*, 2021.
- **Akula, R., & Garibay, I. (2021, July). Ethical AI for social good.** In the *International Conference on Human-Computer Interaction* (pp. 369-380). Cham: Springer International Publishing.

Research Recognition

- NSF I-Corps Entrepreneurial Lead(2019), DARPA Forward Riser (2022), U.S. Patent (2025)

Media Coverage

- Research featured by, NBC News, IEEE Spectrum, Los Angeles Times, Defense One, ScienceDaily, ABC Australia

REFERENCES

1. **Dr. Mauricio Rojas, MD**
Professor and Vice Chair of Research in the Department of Internal Medicine, The Ohio State University
2. **Dr. Ivan Garibay, Ph.D.**
Professor and Director of Complex Systems Research Lab, University of Central Florida
3. **Dr. Damla Turgut, Ph.D.**
Pegasus Professor and chair of the UCF Department of Computer Science, University of Central Florida